Shaharukh Khan

shaharukhkhan4350@gmail.com | +91-7275564486 | Homepage | LinkedIn | GitHub

Education

Indian Institute of Technology (IIT) Kanpur, BT-MT 2013 – 2018

Experience

Applied Scientist-II, Amazon -- Bangalore Nov 2025 – Present

- Trained and deployed a runtime policy violation detection model to safeguard Alexa user interactions
- Deployed a decoder classification model (**Qwen3-Guard**) for **Alexa+** (en_US) with **<40ms** latency and **95% precision**; scaling to global markets
- Built an offline detection module automating retraining pipelines to maintain continuous model performance
- Architected an automated agent generating and annotating policy-aligned synthetic data

Data Scientist-II, Ola Krutrim -- Bangalore Mar 2024 – Oct 2025

- Built **Chitrarth**, a multilingual vision-language model supporting **10+ Indic languages**
- Developed **BharatBench**, a multimodal benchmark evaluating **Indic language understanding**
- Fine-tuned **VideoLLaMA3** and **Chitrarth** for high-accuracy **CCTV video understanding**
- Trained a lightweight image classifier detecting **image orientation** across 4 discrete angles
- Trained **Chitrarth-vl2**, an Indic VLM for 22 languages supporting **OCR**, **JSON output**, and video understanding
- Developed the **MCP agents** orchestration module for the **Kruti application**
- Deployed a **fraud detection system** identifying spare part fraud across **Ola Electric** service centers
- Deployed a **ResNet50 NSFW classification model** for content moderation in the **Kruti application**
- Fine-tuned **Phi-3.5** and **Qwen2.5-VL** for **document intelligence** powering **Ola Cabs** document verification
- Trained a bilingual English-Indic **tokenizer** achieving state-of-the-art compression via **SuperBPE**
- Built **2-tower** retrieval and **DLRM** ranking pipelines for **Ola Foods** and **Groceries**

Data Scientist, Arcana -- Bangalore Aug 2023 – Mar 2024

- Generated **sentiment scores** for earnings call transcripts using **FinBERT**
- Analyzed market factors across commodities, ETFs, options, currencies, momentum, beta, and treasury rates
- Built a **factor-based financial model** achieving an **R-squared of 38%** for asset return prediction

Data Scientist-II, HiLabs Inc. -- Bangalore Nov 2020 – Aug 2023

- Built **DRG prediction models** combining **BERT** medical sentiment with **YOLOv5** chart extraction
- Led a **10+ member team** to develop an AI-based **provider roster ingestion system**
- Managed a **5+ member team** to build **ELIXIR-P**, an **anomaly detection** and remediation engine using Spring Boot, RDBMS, Hadoop, Spark, and Solr
- Received **Xtra Miler** award for outstanding performance

Assistant Manager, HSBC -- Bangalore Jul 2018 – Nov 2020

- Built **Alteryx workflows** for **data quality rules** and developed a **PII detection tool**
- Developed QBot, a TF-IDF **data governance and benchmarking chatbot**; received the **Team Star** award

Publications

IndicVisionBench: Benchmarking Cultural and Multilingual Understanding in VLMs ICLR 2026

BhashaKritika: Building Synthetic Pretraining Data at Scale for Indic Languages AAAI 2026

PRISM: Distilling LLM Reasoning into Compact Guardrail Models via Auxiliary Head Regularization AMLC 2026
Amazon Machine Learning Conference (AMLC 2026)

SAGE: Hierarchical Content Safety via Prefill Readout of Decoder Guard Models AMLC 2026
Amazon Machine Learning Conference (AMLC 2026 Workshop)

Chitrakshara: A Large Multilingual Multimodal Dataset for Indian languages CVPR 2025
VLMs-4-All

Chitrarth: Bridging Vision and Language for a Billion People NeurIPS 2024
Multimodal Algorithmic Reasoning Workshop

Chitranuvad: Adapting Multi-lingual LLMs for Multimodal Translation EMNLP 2024
Workshop on Machine Translation

Seeing Straight: Document Orientation Detection for Efficient OCR arXiv 2025

Technologies

Highly Proficient: Python, PyTorch, Transformers, Hugging Face, LLM, VLM, Fine-tuning, SQL, FastAPI
Familiar: RecSys, Docker, vLLM, RAG, MCP, DPO, RLHF, GRPO, LangGraph, C++, R, PySpark